Comprehensive MES / APS / WMS Solution Proposal

Solar Manufacturing Digital Transformation

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1. Executive Summary

This proposal presents an AI■native Manufacturing Execution System (MES), Advanced Planning & Scheduling (APS), and Warehouse Management System (WMS) tailored for Solar panel manufacturing. The solution addresses current pain points – lack of production tracking and scheduling, no quality traceability, inadequate machine integration, absence of real-time OEE dashboards – and delivers real■time production visibility, intelligent scheduling, ALMM■compliant traceability, and dispatch optimisation.

2. Customer Requirements

Industry	Solar
	solar panels
Production Capacity	300 MW annually
	4
Machine Types	20
	SAP
Current Systems	manual tracking in spreadsheets
	High

Pain Points:

• lack of production tracking and scheduling

• no quality traceability

• inadequate machine integration

• absence of real-time OEE dashboards

Functional Requirements:

#	Requirement

1	production tracking and scheduling
2	quality traceability (batch genealogy from wafer to module)
3	OPC UA machine integration
4	real-time OEE dashboards
5	

3. Industry Best Practices & Recommendations

Recommendations for Solar Manufacturing

1. Recommended MES Modules
Given the specific needs of solar panel manufacturing, the following MES (Manufacturing Execution System) modules are recommended:

- Production Tracking and Scheduling Module: To manage production orders, work orders, and scheduling in real-time.
- Quality Management Module: For tracking quality metrics such as First Pass Yield (FPY), EL testing results, and warranty claims analysis.
- Machine Integration Module: To connect with various machines using OPC UA, Modbus, REST protocols for data collection and control.
- Real-Time OEE Dashboard Module: To monitor Overall Equipment Effectiveness in real-time across all production lines.
- Batch Genealogy Management Module: For tracking materials from wafer to module, ensuring traceability throughout the manufacturing process.

2. Key Process Steps
1. Initial Assessment:
- Conduct a thorough assessment of current systems and processes.
- Identify pain points and areas for improvement in production tracking, scheduling, quality management, and machine integration.

2. MES Configuration:
- Configure MES modules to align with specific solar manufacturing requirements (e.g., lot-size constraints, batch genealogy).
- Set up work center capacity models and scheduling algorithms based on historical data.

3. Machine Integration:
- Implement OPC UA retrofit proof-of-concepts for machines lacking connectivity.
- Ensure secure integration using protocols like Modbus and REST.

4. Training and Change Management:
- Provide comprehensive training to 60 operators on the new MES system.
- Develop a change management plan to address resistance patterns such as "too much data entry" or "machine cannot connect."

5. Deployment and Parallel Run:
- Deploy Gantt dashboards for visualizing production schedules.
- Conduct parallel runs of PODISE vs. manual planning to demonstrate the benefits.

6. Post-Go-Live Support:
- Offer ongoing support

4. MES Functional Scope

The solution covers the entire manufacturing lifecycle, from order receipt to dispatch. Below is a summary of the key modules and their solar■specific capabilities.

Module	Description
Order Management	Import sales orders from SAP, prioritise (govt vs commercial), manage delivery dates, generate production batches.
Production Planning (Projection)	Weekly demand projection, capacity simulation, line allocation, material requirement planning (MRP).
Order Scheduler	Allocate orders to production lines, prioritise urgent/tender orders, minimise changeover time, provide suggested start dates.
Line Scheduler	Finite■capacity scheduling per work centre; batch■aware (laminator: 12, flash tester: 1, packaging: 30); real■time Gantt; automatic re■optimisation on disruptions.
Production Tracking	Real■time OEE, machine status, production counts; integration via OPC UA, MQTT, Modbus; scrap alerts and cycle■time monitoring.
Quality Management	AI■AOI inspection, EL image classification, flash test capture, material genealogy (ingot→wafer→cell→module), ALMM compliance dashboards.
Warehouse Management (WMS)	Raw material, WIP, finished goods inventory; LPN scanning, location tracking, FEFO for EVA/backsheet expiry, dock■to■dock visibility.
Dispatch Planning	Pallet grouping, route■based loading, digital packing list, ALMM Certificate of Compliance (COC) auto■generation.
Truck Scheduling	Optimises dock allocation and truck arrival times to reduce waiting, improve fill rates, and cut logistics cost.

5. Advanced Planning & Scheduling (PODISE/APS)

Traditional SAP MRP cannot handle solar■specific lot sizing, half■cut cell batching, or bifacial routing. PODISE fills this gap with real■time, constraint■based scheduling.

Capability	SAP MRP	PODISE/APS
Scheduling granularity	Day■level	Hour / minute level
Capacity handling	Infinite capacity	Finite capacity

Rush order response	Batch run (1+ hour)	< 30 seconds automated
Lot■size constraints	Basic min/max	Multi■level (solar■specific)
Technology variants	Limited BOM	Full PERC/TOPCon/Bifacial/HJT
Bottleneck visibility	None	Real■time identification
Re■optimisation on breakdown	Manual replanning	Automatic on Kafka event
Gantt visualisation	None	Real■time Angular Gantt (drag■drop)
AI recommendations	None	Plain■English rescheduling advice

Example: When laminator #2 breaks down, PODISE re■optimises all affected orders within 30 seconds, reassigning batches to other laminators and providing an updated delivery forecast to SAP and the Gantt dashboard.

6. System Architecture



1. Architecture Overview

The proposed system architecture for the solar panel factory is designed to enhance production tracking, scheduling, quality management, and machine integration through an integrated MES platform. This architecture leverages advanced technologies such as OPC-UA, Kafka event bus, and a data lake to ensure real-time visibility and control over manufacturing processes while maintaining high standards of security and scalability.

2. System Components

- SAP ERP (SD/PP modules): Manages sales orders, production planning, and procurement.
- Solar MES Platform: Includes Production Order Management, Sales Force Data Collection (SFDC), Quality Management System (QMS), Equipment Integration, and Overall Equipment Effectiveness (OEE) monitoring.
- PODISE Engine: Converts real-time sales orders to production orders, supports lot splitting, and enables

backward/forward scheduling.
- Line Scheduler: Provides batch-aware finite-capacity scheduling with Gantt chart visualization for real-time adjustments.
- Production Tracking Agent PDI/PDO: Collects data from machines using OPC-UA, MQTT, TCP/IP protocols, ingests Excel files via Kafka to PostgreSQL database.
- Quality Inspection Agent: Utilizes AI-based Automated Optical Inspection (AOI), EL image classification, and flash-test capture for quality assurance.
- Predictive Maintenance Agent: Analyzes sensor streams to predict equipment failures before they occur.
- OPC-UA Gateway: Facilitates communication between m

For a detailed diagram, refer to the High■Level Design (HLD) document.

7. Integration Architecture

Integration	Direction	Protocol	Frequency	Data Exchanged
SAP ERP ↔ MES	Bidirectional	REST API / IDoc	Real■time / Hourly	Sales orders, BOM, routings, confirmations, quality results
MES ↔ OPC UA Gateway	Bidirectional	OPC UA	1■5 seconds	Machine commands, sensor data, status, alarms
OPC UA Gateway → Kafka	Unidirectional	OPC UA subscription	Continuous	Timestamped tag values
Kafka → PostgreSQL / InfluxDB	Unidirectional	Kafka Consumer	Continuous	PDI/PDO records, sensor time■series
MES ↔ WMS	Bidirectional	REST API	Event■driven	Pick instructions, inventory confirmations, LPN data
MES → ALMM Portal	Unidirectional	REST API (MNRE)	On dispatch	Module serial, test results, COC
Quality Inspection Agent → PostgreSQL	Unidirectional	Kafka / REST	Every test cycle	EL classification, flash test IV curves, defect codes

8. AI■Native Multi■Agent System

Nine specialised AI agents coordinate via Kafka and MCP tools, powered by on■premises Ollama + Llama 3.2. This ensures data sovereignty and intelligent automation at every step.

Agent	Role	AI Capability
Orchestrator	Goal decomposition, workflow coordination	Fault■tolerant task sequencing
PODISE / APS	Sales order → production order; lot splitting; scheduling	Constraint■based optimisation; real■time re■planning
PDI/PDO Tracking	Machine data ingestion (OPC UA, MQTT, Excel)	Scrap analysis, cycle■time anomaly detection
Quality Inspection	AI■AOI + EL classification + flash test capture	CNN defect detection >98% accuracy

Predictive Maintenance	Sensor stream analysis, failure forecast	MTBF prediction, PM work order creation
Dispatch Planning	AI■optimised truck loading, dock assignment	Route optimisation, fill■rate maximisation
Reporting	Shift■end summaries, plain■English AI narration	LLM■generated management reports

9. Implementation Phases & Roadmap

Phase 1 – Foundation (Weeks 1■8): AS■IS mapping, GAP analysis, blueprint sign■off, MES core configuration, basic SFDC.

Phase 2 – Planning & APS (Weeks 9■20): PODISE/APS deployment, line scheduler, Gantt dashboard, SAP integration.

Phase 3 – Warehouse & Dispatch (Weeks 16■28): WMS go■live (inbound, inventory, outbound), dispatch planning, ALMM COC.

Phase 4 – Equipment Integration & AI (Weeks 12■32): OPC UA connectivity, PDI/PDO agents, AI–AOI, predictive maintenance.

Phase 5 – Testing & Go■Live (Weeks 28■36): UAT, parallel run, operator training, go■live (line■by■line), hypercare.

Phase 6 – Optimisation & Expansion (Weeks 36■52): KPI monitoring, AI model tuning, remaining line rollout, ALMM compliance automation.

10. Effort & Cost Estimation

Total Effort (person■months)	88.5
	20.5
Estimated Cost (USD)	$752,250

Phase Breakdown:

Phase	Duration
Discovery & Blueprint	1.5 months
Core MES Configuration	6.0 months
Equipment Integration	10.0 months
Testing & Go-Live	2.0 months
Post Go-Live Support	1.0 months

11. ROI & Business Case

Metric	Value
	$752,250
Annual Savings	$1,504,500
	24 months
5■Year Net Benefit	$6,770,250
	15%
OEE Improvement	8%
	10%

Industry Benchmarks:

Based on the provided requirements for a solar manufacturing plant with an annual production capacity of 300 MW, here are the relevant financial benchmarks and payback periods:

Financial Benchmarks and Payback Periods

- Production Capacity: 200-500 MW (Mid-size plant)
- Typical MES Payback Period:
- 12-18 months

Given that your production capacity falls within the mid-size range, you can expect a payback period of approximately 12 to 18 months. This aligns with your expectation for ROI within two years.

Cost Reduction Potentials and Efficiency Gains

Cost Reduction:
- Manual Tracking in Spreadsheets: Transitioning from manual tracking to an automated MES system will reduce labor costs associated with data entry, error correction, and manual scheduling.
- Estimated reduction: ~5x less data entry compared to Excel-based systems.

- Quality Traceability: Implementing quality traceability can significantly reduce rework and scrap rates by identifying issues early in the production process.
- Potential savings: Up to 10% of annual production costs through reduced waste.

Efficiency Gains:
- Production Tracking & Scheduling: Real-time scheduling and tracking will improve over

12. Commercial Model

License Model: Annual subscription or perpetual license with annual maintenance (AMC).
Implementation Services: Fixed■price phased approach; billed per phase upon milestone completion.
Team Structure: Program Manager, Solution Architect, MES Consultant, APS Expert, Integration Engineer, QA Lead.
Support: 24×7 hypercare for 4 weeks post go■live; optional extended support SLA.

Infrastructure: On■premise servers, GPU for AI, edge gateways – detailed in HLD.
Training: Hands■on operator training at workstations, supervisor dashboards, management analytics.

13. Risk Assessment

A comprehensive Risk Assessment Report is provided as a separate document. The top risks include integration complexity, machine data availability, user adoption, and schedule slippage. Mitigation plans are detailed in the risk register.

14. Next Steps

1. Customer reviews this proposal and provides written acceptance.
2. Kick■off meeting within 2 weeks of acceptance.
3. Phase 1 (Discovery & Blueprint) commences with dedicated team.
4. Detailed project plan, RACI matrix, and risk register delivered by end of Phase 1.
5. Pilot line go■live (MES + APS) within 20 weeks; subsequent lines rolled out iteratively.
